

ANALYSIS

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1. CONVERGENT SEQUENCES

Definition 1.1 (Convergent Sequence). A sequence $\langle s_n \rangle$ in M converges to $l \in M$, if

$$(1.1) \quad \forall \varepsilon > 0 : \exists N \in \mathbb{N} : \forall n > N : d(s_n, l) < \varepsilon$$

Theorem 1.2. Every convergent sequence is bounded.

Proof. Let $M = (A, d)$ be a metric space and let $\langle s_n \rangle$ be a sequence in M that converges to $l \in M$. Let $\varepsilon > 0$. Since $\langle s_n \rangle$ converges, there is an N such that if $n > N$

$$(1.2) \quad d(s_n, l) < \varepsilon$$

Set B' be a bound of the sequence before N

$$B' := \max\{d(s_n, l) : n \leq N\}$$

Finally set

$$(1.3) \quad B := \max\{B', \varepsilon\}$$

Then B is a bound for all $n \in \mathbb{N}$

$$(1.4) \quad \forall n \in \mathbb{N} : d(s_n, l) \leq B$$

which implies that $\langle s_n \rangle$ is bounded. □

Theorem 1.3. If

$$(1.5) \quad \begin{aligned} S &= \lim_n s_n \\ T &= \lim_n t_n \end{aligned}$$

Then

$$(1.6) \quad \lim_n (s_n + t_n) = S + T$$

Proof. Choose an $\varepsilon' > 0$ and define $\varepsilon := 2\varepsilon'$. By the definition of convergence, there exists an N_s and N_t such that

$$(1.7) \quad n > N_s \Rightarrow |s_n - S| < \varepsilon'$$

$$(1.8) \quad n > N_t \Rightarrow |t_n - T| < \varepsilon'$$

Let $N := \max\{N_s, N_t\}$, then for $n > N$, by the triangle inequality

$$(1.9) \quad |(s_n + t_n) - (S + T)| \leq |s_n - S| + |t_n - T|$$

By the definition of convergence

$$(1.10) \quad |s_n - S| + |t_n - T| < \varepsilon' + \varepsilon' = \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

Thus

$$(1.11) \quad \lim_n (s_n + t_n) = S + T$$

□

Theorem 1.4. *If*

$$(1.12) \quad \begin{aligned} S &= \lim_n s_n \\ T &= \lim_n t_n \end{aligned}$$

Then

$$(1.13) \quad \lim_n (s_n \cdot t_n) = ST$$

Proof. Choose an $\varepsilon' > 0$. By the definition of convergence, there exists an N_s and N_t such that

$$(1.14) \quad n > N_s \Rightarrow |s_n - S| < \varepsilon'$$

$$(1.15) \quad n > N_t \Rightarrow |t_n - T| < \varepsilon'$$

Let $N := \max\{N_s, N_t\}$, then for $n > N$

$$(1.16) \quad |(s_n t_n) - (ST)| = |s_n t_n - s_n T + s_n T - ST|$$

$$(1.17) \quad = |s_n(t_n - T) + (s_n - S)T|$$

By the triangle inequality

$$(1.18) \quad \begin{aligned} |s_n(t_n - T) + (s_n - S)T| &\leq |s_n(t_n - T)| + |(s_n - S)T| \\ &\leq |s_n||t_n - T| + |s_n - S||T| \end{aligned}$$

Since every convergent sequence is also bounded, there is some bound B , such that for every n

$$(1.19) \quad |s_n| < B$$

$$(1.20) \quad |s_n||t_n - T| + |s_n - S||T| \leq B|t_n - T| + |s_n - S||T|$$

Let $M := \max\{B, |T|\}$. Then

$$(1.21) \quad B|t_n - T| + |s_n - S||T| \leq M(|s_n - S| + |t_n - T|)$$

Let $\varepsilon := 2M\varepsilon'$, then

$$(1.22) \quad |s_n - S| < \varepsilon' = \frac{\varepsilon}{2M}$$

$$(1.23) \quad |t_n - T| < \varepsilon' = \frac{\varepsilon}{2M}$$

Then

$$(1.24) \quad M(|s_n - S| + |t_n - T|) < M\left(\frac{\varepsilon}{2M} + \frac{\varepsilon}{2M}\right) = \varepsilon$$

Thus, for $n > N$

$$(1.25) \quad |s_n t_n - ST| < \varepsilon$$

or

$$(1.26) \quad \lim_n s_n t_n = ST$$

□

Theorem 1.5. *If a sequence converges, then the limit is unique.*

Proof. Assume without loss of generality that some convergent sequence s has two distinct limits L and L' . Define the distance between these two limits $d := |L - L'|$. From the definition of the metric, $d > 0$. From the definition of convergence, there is a N and N' such that

$$(1.27) \quad n > N \Rightarrow |s_n - L| < \frac{d}{2}$$

and

$$(1.28) \quad n > N' \Rightarrow |s_n - L'| < \frac{d}{2}$$

Let $M := \max\{N, N'\}$. For $n > M$

$$(1.29) \quad d := |L - L'| = |L - s_n + s_n - L'|$$

By the triangle identity

$$(1.30) \quad |L - s_n + s_n - L'| \leq |s_n - L| + |s_n - L'|$$

From the definition of convergence

$$(1.31) \quad |s_n - L| < \frac{d}{2}$$

$$(1.32) \quad |s_n - L'| < \frac{d}{2}$$

By $a < b \wedge c < d \Rightarrow a + c < b + d$

$$(1.33) \quad |s_n - L| + |s_n - L'| < \frac{d}{2} + \frac{d}{2} = d$$

which implies

$$(1.34) \quad d < d$$

which is a contradiction. $\square$

Theorem 1.6 (Squeeze Theorem). *For all sequences s , and all convergent sequences r, t such that*

$$(1.35) \quad \begin{aligned} \lim_n r_n &= L \\ \lim_n t_n &= L \end{aligned}$$

and for all $n \in \mathbb{Z}^+$

$$(1.36) \quad r_n < s_n < t_n$$

implies that s is a convergent sequence and its limit is given by

$$(1.37) \quad \lim_n s_n = L$$

Proof. Fix $\varepsilon > 0$. Then by the definition of the convergent sequence

$$(1.38) \quad \begin{aligned} \exists N_1 : n > N_1 &\Rightarrow |y_n - L| < \varepsilon \\ \exists N_2 : n > N_2 &\Rightarrow |z_n - L| < \varepsilon \end{aligned}$$

Let $N := \max\{N_1, N_2\}$, then for all $n > N$

$$(1.39) \quad \begin{aligned} L - \varepsilon &< y_n < L + \varepsilon \\ L - \varepsilon &< z_n < L + \varepsilon \end{aligned}$$

Using the trichotomy property of order

$$(1.40) \quad L - \varepsilon < y_n < x_n < z_n < L + \varepsilon$$

or

$$(1.41) \quad L - \varepsilon < x_n < L + \varepsilon$$

Therefor $n > N$ implies

$$(1.42) \quad |x_n - L| < \varepsilon$$

or

$$(1.43) \quad \lim_n x_n = L$$

□

Theorem 1.7.

$$(1.44) \quad \lim_n \frac{1}{n} = 0$$

Proof. Choose an $\varepsilon' \in \mathbb{R}^+$. By the Archimedean property

$$(1.45) \quad \exists N \in \mathbb{Z}^+ : N > \varepsilon'$$

Since both $N > 0$ and $\varepsilon' > 0$

$$(1.46) \quad \frac{1}{N} < \frac{1}{\varepsilon'}$$

Define $\varepsilon := \frac{1}{\varepsilon'} \in \mathbb{R}^+$

$$(1.47) \quad \frac{1}{N} < \varepsilon$$

Let $n > N > 0$, then

$$(1.48) \quad \frac{1}{N} > \frac{1}{n}$$

By trichotomy

$$(1.49) \quad \frac{1}{n} < \varepsilon$$

$$(1.50) \quad n > N \Rightarrow \left| \frac{1}{n} - 0 \right| < \varepsilon$$

or

$$(1.51) \quad \lim_n \frac{1}{n} = 0$$

□

Lemma 1.8. *If $\{|a_n - L|\}$ converges to zero, then $\{a_n\}$ converges to L .*

$$(1.52) \quad \lim_n |a_n - L| = 0 \Rightarrow \lim_n a_n = L$$

Proof. Since $\{|a_n - L|\}$ converges to zero, for any $\varepsilon > 0$, there is an N

$$(1.53) \quad n > N \Rightarrow ||a_n - L| - 0| < \varepsilon$$

$$(1.54) \quad ||a_n - L| < \varepsilon$$

$$(1.55) \quad n > N \Rightarrow |a_n - L| < \varepsilon$$

Then

$$(1.56) \quad \lim_n a_n = L$$

□

2. CLOSED SETS

Definition 2.1 (Closed Set). A subset $S \subset M$ is *closed* in M if every sequence $\langle s_n \rangle$ in S that converges in M also converges in S .

$$(2.1) \quad \lim_n s_n \in M \Rightarrow \lim_n s_n \in S$$

Theorem 2.2. *The real interval $[a, b]$ is closed.*

$$(2.2) \quad \forall s : \mathbb{N} \rightarrow [a, b] : \lim_n s_n = L \Rightarrow L \in [a, b]$$

Proof. Suppose not, then

$$(2.3) \quad \exists s : \mathbb{N} \rightarrow [a, b] : \lim_n s_n = L \wedge (L < a \vee b < L)$$

From the definition of a limit

$$(2.4) \quad \forall \varepsilon > 0 : \exists N \in \mathbb{N} : n > N \Rightarrow L - \varepsilon < s_n < L + \varepsilon$$

Case 1 ($b < L$). Then $L - b > 0$. Let $\varepsilon = L - b$, then there is a N_1 , such that $n > N_1$

$$(2.5) \quad L - L + b = b < s_n$$

but the definition of the sequence restricts the domain to $s_n < b$, so we obtain a contradiction.

Case 2 ($a < L$). Then $a - L > 0$, let $\varepsilon = a - L$, which implies there is a N_2 , such that $n > N_2$

$$(2.6) \quad s_n < L + a - L = a$$

which contradicts the restriction on the domain $a < s_n$

□

Theorem 2.3 (Inheritance Principle). *If S is closed in M and N is a subspace of M , then $N \cap S$ is closed in N .*

Proof. Let N be a subspace of M and S some closed subset of M . Define a sequence $s : \mathbb{N} \cap S$ such that

$$(2.7) \quad \exists L \in N : \forall \varepsilon > 0 : \exists N \in \mathbb{N} : \forall n \in \mathbb{N} : |s_{N+n} - L| < \varepsilon$$

Since S is closed and $N \cap M$, $L \in S \cap N$. Hence $S \cap N$ is closed.

□

Corollary 2.4. *If N is closed in M , then S is closed in N if and only if S is closed in M .*

3. MONOTONIC SEQUENCES

Definition 3.1 (Monotonic Sequence). An *increasing sequence* is a sequence where

$$(3.1) \quad m > n \Rightarrow s_m > s_n$$

A *strictly increasing sequence* is a sequence where

$$(3.2) \quad m > n \Rightarrow s_m \geq s_n$$

Similarly, a *decreasing sequence* is a sequence where

$$(3.3) \quad m > n \Rightarrow s_m < s_n$$

And a *strictly decreasing sequence* is a sequence where

$$(3.4) \quad m > n \Rightarrow s_m \leq s_n$$

A *monotonic sequence* is a sequence that is either increasing or decreasing.

Lemma 3.2. If $n : \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ is a strictly increasing sequence ($i > j \Rightarrow n_i > n_j$), then for any positive integer N

$$(3.5) \quad \exists i \in \mathbb{Z}^+ : n_i > N$$

Proof. Hypothesis for induction:

$$(3.6) \quad \forall i \in \mathbb{Z}^+ : n_i \geq i$$

Case 1 (Base). If $n_1 \in \mathbb{Z}^+$, then $n_1 \geq 1$.

Case 2 (Induction). Assume without loss of generality,

$$(3.7) \quad n_i \geq i$$

Since n is strictly increasing

$$(3.8) \quad n_{i+1} > n_i$$

By trichotomy

$$(3.9) \quad n_{i+1} > i$$

Then

$$(3.10) \quad n_{i+1} \geq i + 1$$

Thus by induction,

$$(3.11) \quad \forall i \in \mathbb{Z}^+ : n_i \geq i$$

□

4. SUBSEQUENTIAL LIMITS

Definition 4.1 (Subsequence). A *subsequence* of a sequence s is a sequence that is a composition $s \circ n$ where $n : \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ is strictly increasing.

Definition 4.2 (Tail). The N -tail of a sequence $s : n \mapsto s_n$ is the subsequence s_{N+n} , where $N \in \mathbb{Z}^{\geq 0}$

Theorem 4.3. If a sequence converges to a limit, then all of its subsequential limits are equal to that limit.

Proof. Consider a sequence s , that converges to L , and $s \circ n$ a subsequence of s . Fix $\varepsilon > 0$. Since s converges to L , there exists an N such that

$$(4.1) \quad n > N \Rightarrow |s_n - L| < \varepsilon$$

Let $I := \inf\{i \in \mathbb{Z}^+ : n_i > N\}$ (The set is not empty, by the lemma above $n_N \geq N$). Since $n_I \geq I > N$

$$(4.2) \quad n > n_I \Rightarrow |s_n - L| < \varepsilon$$

From the definition of strictly increasing

$$(4.3) \quad i > I \Rightarrow n_i > n_I$$

So

$$(4.4) \quad i > I \Rightarrow n_i > n_I \Rightarrow |s_{n_i} - L| < \varepsilon$$

Or

$$(4.5) \quad \lim_i s_{n_i} = L$$

□

Theorem 4.4. *If a sequence is bounded, then all subsequences have the same bounds.*

Proof. Let s be a sequence bounded by (b, B)

$$(4.6) \quad \forall n : b \leq s_n \leq B$$

Let $s \circ n$ be some subsequence, then

$$(4.7) \quad \forall i : b \leq s_{n_i} \leq B$$

□

Theorem 4.5. *Every subsequence of a monotonic sequence is the same type of monotonic.*

Proof. Let s be a sequence and $\sim$ be some order relation on the monotonic sequence such that

$$(4.8) \quad m > n \Rightarrow s_m \sim s_n$$

Let $s \circ n$ be some subsequence, then

$$(4.9) \quad i > j \Rightarrow n_i > n_j \Rightarrow s_{n_i} \sim s_{n_j}$$

□

Lemma 4.6. *If a tail of a sequence has a maximum, the parent sequence has a maximum.*

Proof. Let $a := \max\{s_n : n > N\}$, which exists by assumption. Let $b := \max\{s_n : n \leq N\}$, which exists because the set is finite. If we let $c := \max\{a, b\}$, then $c = \max\{s_n : n \in \mathbb{Z}^+\}$. □

Theorem 4.7 (Point Peak Theorem). *Every real sequence has a monotonic subsequence.*

Proof. Let $\langle s_n \rangle$ be any real sequence. Either every tail of $\langle s_n \rangle$ has a maximum

$$(4.10) \quad \forall N \in \mathbb{Z}^{\geq 0}, \exists m, \forall n : s_{N+m} \geq s_{N+n}$$

or there is a tail without a maximum.

$$(4.11) \quad \exists N \in \mathbb{Z}^{\geq 0}, \forall m, \exists n : s_{N+m} < s_{N+n}$$

First suppose every tail of $\langle s \rangle$ has a maximum

$$\forall N \in \mathbb{Z}^{\geq 0}, \exists m, \forall n : s_{N+m} \geq s_{N+n}$$

Define a sequence $k : \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$

$$(4.12) \quad k_1 := \min\{m \mid \forall n : s_m \geq s_n\}$$

$$(4.13) \quad k_{i+1} := \min\{m \mid \forall n : s_{k_i+m} \geq s_{k_i+n}\} + k_i$$

The sequence $s \circ k$ is a subsequence because k strictly increasing

$$(4.14) \quad m \in \mathbb{Z}^+ \Rightarrow k_{i+1} > k_i$$

From the definition of k_{i+1}

$$(4.15) \quad \forall i, n : s_{k_{i+1}} \geq s_{k_i+n}$$

Since k is strictly increasing

$$(4.16) \quad k_{i+2} > k_{i+1} > k_i$$

Let n be the positive integer such that

$$(4.17) \quad n := k_{i+2} - k_i \in \mathbb{Z}^+$$

Applying this n to the definition

$$(4.18) \quad s_{k_{i+1}} \geq s_{k_i+k_{i+2}-k_i} = s_{k_{i+2}}$$

which shows that $s \circ k$ is decreasing.

$$(4.19) \quad s_{k_i} \geq s_{k_{i+1}}$$

Now suppose there is a tail that does not have a maximum

$$\exists N \in \mathbb{Z}^{\geq 0}, \forall m, \exists n : s_{N+m} < s_{N+n}$$

Every tail of the tail without a maximum has no maximum, by lemma 6.6.

$$(4.20) \quad \forall M \in \mathbb{Z}^{\geq 0}, m, \exists n : s_{N+M+m} < s_{N+M+n}$$

This implies there is always a larger number further down the sequence.

$$(4.21) \quad \forall m, \exists n : s_{N+m} < s_{N+m+n}$$

Define a sequence $k : \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$

$$(4.22) \quad k_1 := N + 1$$

$$(4.23) \quad k_{i+1} := \min\{n : s_{k_i+n} > s_{k_i}\} + k_i$$

The sequence $s \circ k$ is a subsequence because k is strictly increasing

$$(4.24) \quad n \in \mathbb{Z}^+ \Rightarrow k_{i+1} > k_i$$

From the definition of k_{i+1} , $s \circ k$ is strictly increasing

$$(4.25) \quad s_{k_{i+1}} > s_{k_i}$$

Thus all real sequences have monotonic subsequences. $\square$

Theorem 4.8. For $|a| < 1$

$$(4.26) \quad \lim_n a^n = 0$$

Proof. Let a be a number such that $|a| < 1$. Since $|a| > 0$, multiplying the inequalities by $|a|$ gives

$$(4.27) \quad 0 \geq |a|^2 < |a|$$

$$(4.28) \quad 0 \geq |a^2| < |a|$$

Assume without loss of generality

$$(4.29) \quad 0 \geq |a^{n+1}| < |a^n|$$

Then multiplying by $|a|$ gives

$$(4.30) \quad 0 \geq |a^{n+1}| |a| < |a^n| |a|$$

$$(4.31) \quad 0 \geq |a^{n+2}| < |a^{n+1}|$$

Then, by induction, for all $n \in \mathbb{Z}^+$

$$(4.32) \quad 0 \geq |a^{n+1}| < |a^n|$$

which implies that the sequence $|a^n|$ is strictly decreasing, and bounded below by zero. By the monotonic convergence theorem, $|a^n|$ converges. Assume it converges to limit L .

$$(4.33) \quad L := \lim_n |a^n|$$

The sequence $|a^{2n}|$ is a subsequence of $|a^n|$, so it converges to the same limit

$$(4.34) \quad L = \lim_n |a^{2n}|$$

$$(4.35) \quad = \lim_n |a^n|^2$$

$$(4.36) \quad = \left(\lim_n |a^n| \right)^2$$

$$(4.37) \quad = L^2$$

Thus, either $L = 0$ or $L = 1$. Since $|a^1| = |a| < 1$ and a^n is strictly decreasing, $L \neq 1$. Therefore

$$(4.38) \quad \lim_n |a^n| = 0$$

By lemma 3.8

$$(4.39) \quad \lim_n a^n = 0$$

□

5. COMPACT SETS

Definition 5.1 (Compact Set). A set is *compact* if every sequence $s : \mathbb{Z}^+ \rightarrow S$ has a subsequential limit in S .

Theorem 5.2. A compact subset of M is a closed subset of M

Proof. Let S be a compact subset of M and $s : \mathbb{N} \rightarrow S$ be a sequence that converges to some $L \in M$. Since s converges, every subsequential limit is equal to L . Since S is compact $L \in S$. Thus S is closed. □

Theorem 5.3. *A compact subset of M is bounded in M*

Proof. Let p be a point in M and $S \subset M$. Suppose S is not bounded, then there is some sequence $s : \mathbb{N} \rightarrow S$ defined by

$$(5.1) \quad d(s_n, p) \geq n$$

Since S is compact, there is some subsequence of s that converges. For every $\varepsilon > 0$, there is an N

$$(5.2) \quad i > N \Rightarrow d(s_{n_i}, p) < \varepsilon$$

Choose some $\varepsilon > 0$. By the Archimedean property there is an N' such that

$$(5.3) \quad \varepsilon < N'$$

Let $M := \max\{N, N'\}$. Since $M > N$

$$(5.4) \quad d(s_{n_M}, p) < \varepsilon < N' < M$$

And since $s \circ n$ is a subsequence $M \leq n_M$, which along with the definition of the sequence gives

$$(5.5) \quad M \leq n_M \leq d(s_{n_M}, p) < M$$

which is a contradiction. So S is bounded. $\square$

6. FUNDAMENTAL SEQUENCES

Definition 6.1 (Fundamental Sequence). A sequence is fundamental if every ε has a N -tail, that is ε -pinched.

$$(6.1) \quad \forall \varepsilon > 0 : \exists N \in \mathbb{N} : \forall m, n > N : d(s_m, s_n) < \varepsilon$$

Theorem 6.2. *Every convergent sequence is fundamental.*

7. COMPLETE SPACES

Definition 7.1. A space is complete if every fundamental sequence is convergent.

Theorem 7.2 (Monotone Convergence Theorem). *A monotonic sequence converges if and only if it is bounded.*

Proof. Let s be an increasing sequence bounded above by B .

$$(7.1) \quad m > n \Rightarrow B \geq s_m \geq s_n$$

Without loss of generality, assume s is non-fundamental

$$(7.2) \quad \exists \varepsilon > 0, \forall N : \exists m, n > N : |s_m - s_n| > \varepsilon$$

Let $m > n$ (If $m = n$, then $s_m - s_m = 0 \not> \varepsilon$).

$$(7.3) \quad \forall n : \exists m > n : |s_m - s_n| > \varepsilon$$

Since s is increasing

$$(7.4) \quad \forall n : \exists m > n \Rightarrow s_m - s_n > \varepsilon$$

$$(7.5) \quad s_m > \varepsilon + s_n$$

Let $p := m - n \Rightarrow m = k + n$

$$(7.6) \quad \forall n, \exists p : s_{n+p} > \varepsilon + s_n$$

Define a sequence $k : \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$

$$(7.7) \quad k_1 := 1$$

$$(7.8) \quad k_{i+1} := \min\{p \mid s_{k_i+p} > s_{k_i} + \varepsilon\} + k_i$$

The sequence $s \circ k$ is a subsequence because k is strictly increasing

$$(7.9) \quad p \in \mathbb{Z}^+ \Rightarrow k_{i+1} > k_i$$

By construction of k

$$(7.10) \quad s_{k_{i+1}} > s_{k_i} + \varepsilon$$

Let $A := s_1 = s_{n_1}$. Consider the induction hypothesis

$$(7.11) \quad \forall i > 1 : s_{k_i} - A > (i-1)\varepsilon$$

Case 1 (Base ($i = 2$)).

$$(7.12) \quad s_{n_2} > s_{n_1} + \varepsilon = A + (2-1)\varepsilon$$

Case 2 (Induction). Assume without loss of generality

$$(7.13) \quad s_{k_i} > A + (i-1)\varepsilon$$

Along with

$$(7.14) \quad s_{k_{i+1}} - \varepsilon > s_{k_i}$$

Gives

$$(7.15) \quad s_{k_{i+1}} - \varepsilon > A + (i-1)\varepsilon$$

$$(7.16) \quad s_{k_{i+1}} - A > [(i+1)-1]\varepsilon$$

By induction

$$(7.17) \quad \forall i > 1 : s_{n_i} - A > (i-1)\varepsilon$$

Choose $N \in \mathbb{Z}^+$ such that

$$(7.18) \quad N > \frac{B-A}{\varepsilon} + 1$$

By the Archimedean property, N exists.

$$(7.19) \quad s_N - A > \left(\frac{B-A}{\varepsilon} + 1 - 1 \right) \varepsilon$$

$$(7.20) \quad s_N > B$$

Which contradicts the assumption that s is bounded above by B . Therefor if s is bounded above, it is fundamental. By the completion theorem, since s is fundamental, it is also convergent. Thus it has shown that bounded, increasing sequences are convergent. If s is a bounded, monotonic decreasing sequence, then $-s$ is bounded, monotonic increasing sequence, and therefor convergent, which implies s is convergent. Thus all bounded, monotonic sequences are convergent. The converse is proven by theorem 3.2, all convergent sequences are bounded. $\square$

8. SUMMABLE SEQUENCES

Definition 8.1 (Series). A *series* of partial sums of a is the sequence

$$(8.1) \quad s_n := \sum_{i=1}^n a_i$$

Definition 8.2 (Summable Sequence). A sequence is *summable* if its series of partial sums converges

$$(8.2) \quad \sum_{i=1}^{\infty} a_i := \lim_n \sum_{i=1}^n a_i$$

The *sum* of a sequence is the limit of its series.

Theorem 8.3. *Each sequence with the same sign has a monotonic series.*

- *A non-negative sequence has an increasing series.*
- *A positive sequence has a strictly increasing series.*
- *A non-positive sequence has a decreasing series.*
- *A negative sequence has a strictly decreasing series.*

Proof. Let $\sim$ be the relationship between the sequence a and zero.

$$(8.3) \quad \forall n : a_n \sim 0$$

Then

$$(8.4) \quad a_{n+1} \sim 0$$

$$(8.5) \quad a_{n+1} + s_n \sim s_n$$

$$(8.6) \quad s_{n+1} \sim s_n$$

□

Theorem 8.4 (Series Linearity). *If a and b are summable and c and d are some numbers, then $ca + bd$ is summable and*

$$(8.7) \quad c \sum_{i=1}^{\infty} a_i + d \sum_{i=1}^{\infty} b_i = \sum_{i=1}^{\infty} (ca_i + db_i)$$

Proof.

$$(8.8) \quad c \sum_{i=1}^{\infty} a_i + d \sum_{i=1}^{\infty} b_i = c \lim_n \sum_{i=1}^n a_i + d \lim_n \sum_{i=1}^n b_i$$

$$(8.9) \quad = \lim_n \left(c \sum_{i=1}^n a_i + d \sum_{i=1}^n b_i \right)$$

$$(8.10) \quad = \lim_n \sum_{i=1}^n (ca_i + db_i)$$

$$(8.11) \quad = \sum_{i=1}^{\infty} (ca_i + db_i)$$

□

Theorem 8.5 (Fundamental Criterion). *A sequence a is summable if and only if*

$$(8.12) \quad \forall \varepsilon > 0 : \exists N : n, m > N : \left| \sum_{i=m+1}^n a_i \right| < \varepsilon$$

Proof. Fix $\varepsilon > 0$. By assumption, there is an N such that

$$(8.13) \quad n, m > N \Rightarrow \left| \sum_{i=m+1}^n a_i \right| < \varepsilon$$

If $n \leq m$, $\sum_{i=m+1}^n a_i = 0$, which is less than all $\varepsilon > 0$. Restrict $n > m$, then

$$(8.14) \quad \left| \sum_{i=1}^n a_i - \sum_{i=1}^m a_i \right| < \varepsilon$$

which by the completeness theorem, shows that the series of a converges and a is summable. $\square$

Theorem 8.6 (Vanishing Condition). *If a is summable*

$$(8.15) \quad \lim_n a_n = 0$$

Proof. Since a is summable it has a sum. Let S be the sum of a

$$(8.16) \quad S := \sum_{i=1}^{\infty} a_i$$

Then

$$(8.17) \quad \lim_n a_n = \lim_n \left(\sum_{i=1}^n a_i - \sum_{i=1}^{n-1} a_i \right)$$

$$(8.18) \quad = \lim_n \sum_{i=1}^n a_i - \lim_n \sum_{i=1}^{n-1} a_i$$

$$(8.19) \quad = S - S = 0$$

$\square$

Theorem 8.7 (Boundness Criterion). *A non-negative sequence is summable, if and only if its series is bounded.*

Proof. Let a be a non-negative sequence. Then its series is increasing. By the monotonic convergence theorem, this monotonic series is convergent if and only if it is bounded. Therefor the non-negative sequence is summable if and only if it is bounded. $\square$

Theorem 8.8 (Comparison Test). *If $\forall n : |a_n| \leq b_n$ and b is summable, then a is summable*

Proof. Choose $\varepsilon > 0$. By fundamental criterion, there is an N , such that $m, n > N$

$$(8.20) \quad \left| \sum_{i=m}^n b_i \right| < \varepsilon$$

Since b is non-negative

$$(8.21) \quad \sum_{i=m}^n b_i = \left| \sum_{i=m}^n b_i \right| < \varepsilon$$

Using the triangle inequality

$$(8.22) \quad \left| \sum_{i=m}^n a_i \right| \leq \sum_{i=m}^n |a_i| \leq \sum_{i=m}^n b_i < \varepsilon$$

By the fundamental criterion, a is then summable. $\square$

Theorem 8.9 (Geometric Series). *If $|a| < 1$, then $\langle a^k \rangle$ is summable. The sum is given by*

$$(8.23) \quad \sum_{i=0}^{\infty} a^i = \frac{1}{1-a}$$

Proof. Let $\langle s_n \rangle$ be the series of partial sums

$$(8.24) \quad s_n := \sum_{i=0}^n a^i$$

$$(8.25) \quad \lim_n s_n = \lim_n \frac{1 - a^{n+1}}{1 - a} = \frac{1}{1 - a}$$

$\square$

Lemma 8.10.

$$(8.26) \quad \limsup_n s_n < B \Rightarrow \exists N \in \mathbb{N} : \forall n \geq N : s_n \leq B$$

Proof. Let $\langle s_n \rangle$ be a bounded sequence

$$(8.27) \quad L := \limsup_n s_n$$

Define B such that

$$(8.28) \quad L < B$$

Suppose that

$$(8.29) \quad \forall N \in \mathbb{N} : \exists n > N : s_n > B$$

Define a sequence $k : \mathbb{N} \rightarrow \mathbb{N}$, such that

$$(8.30) \quad k_1 = \min\{n : s_n > B\}$$

$$(8.31) \quad k_{i+1} = \min\{n : s_{k_i+n} > B\} + k_i$$

Since $k_{i+1} > k_i$, $\langle s \circ k \rangle$ is a subsequence. By its definition, $\forall i \in \mathbb{N} : s_{k_i} > B > L$. Then $\lim_i s_{k_i} > \limsup_n s_n$ which is a contradiction. Therefore $\exists N : \forall n \geq N : s_n < B$ $\square$

Theorem 8.11 (Ratio Test). *If*

$$(8.32) \quad \limsup_n \left| \frac{a_{n+1}}{a_n} \right| < 1$$

Then a is summable.

Proof. Let a be some sequence such that

$$(8.33) \quad \limsup_n \left| \frac{a_{n+1}}{a_n} \right| < 1$$

Define s such that

$$(8.34) \quad \limsup_n \left| \frac{a_{n+1}}{a_n} \right| < s < 1$$

By the above lemma, there is an N , such that for every $n \geq N$

$$(8.35) \quad \left| \frac{a_{n+1}}{a_n} \right| < s$$

$$(8.36) \quad |a_{n+1}| < s|a_n|$$

Induction hypothesis

$$(8.37) \quad \forall k \in \mathbb{Z}^+ : |a_{N+k}| < s^k |a_N|$$

The base case is given by $n = N$

$$(8.38) \quad |a_{N+1}| < s|a_N|$$

Assume without loss of generality $|a_{N+k}| < s^k |a_N|$, multiplying by $s > 0$

$$(8.39) \quad |a_{N+k}|s < s^{k+1}|a_N|$$

Applying the inequality from before

$$(8.40) \quad |a_{N+k+1}| < |a_{N+k}|s < s^{k+1}|a_N|$$

Thus, by induction

$$(8.41) \quad \forall k \in \mathbb{Z}^+ : |a_{N+k}| < s^k |a_N|$$

By the comparison test, a_{N+k} is summable, therefor a_n is summable. $\square$

Definition 8.12 (Absolutely Summable). A sequence is *absolutely summable* if $\langle |a_n| \rangle$ is summable.

Theorem 8.13. *An absolutely summable sequence is summable.*

Proof. By assumption the series

$$(8.42) \quad \sum_{i=1}^{\infty} |a_i|$$

converges. Then by the fundamental criterion, for every $\varepsilon > 0$, there is an N , such that

$$(8.43) \quad m, n > N \Rightarrow \left| \sum_{i=m+1}^n |a_i| \right| < \varepsilon$$

$$(8.44) \quad \sum_{i=m+1}^n |a_i| \leq \left| \sum_{i=m+1}^n |a_i| \right| < \varepsilon$$

By the triangle identity

$$(8.45) \quad \left| \sum_{i=m+1}^n a_i \right| \leq \sum_{i=m+1}^n |a_i| < \varepsilon$$

Then, for every $\varepsilon > 0$, there is an N , such that

$$(8.46) \quad m, n > N \Rightarrow \left| \sum_{i=m+1}^n a_i \right|$$

By the fundamental criterion, a is summable. $\square$

Lemma 8.14. *If $\langle a_n \rangle$ is non-negative and decreasing, then*

$$(8.47) \quad \forall m, n \in \mathbb{N} : \left| \sum_{i=m}^n (-1)^i a_i \right| \leq a_m$$

Proof. Fix $m \in \mathbb{N}$. Let the induction hypothesis be that for all $n \in \mathbb{N}$

$$(8.48) \quad 0 \leq \sum_{i=m}^n (-1)^{i-m} a_i \leq a_m$$

Case 1 (Basis for Induction $P(m)$). Since $\langle a_n \rangle$ is non-negative

$$(8.49) \quad 0 \leq a_m = (-1)^{m-m} a_m = \sum_{i=m}^m (-1)^{i-m} a_i$$

Case 2 (Basis for Induction $P(m+1)$). Since $\langle a_n \rangle$ is non-negative

$$\begin{aligned} a_m &\geq -a_{m+1} + a_m \geq 0 \\ a_m &\geq (-1)^{m+1-m} a_{m+1} + (-1)^{m-m} a_m \geq 0 \\ a_m &\geq \sum_{i=m}^{m+1} (-1)^{i-m} a_i \geq 0 \end{aligned}$$

For the induction step, suppose for $\forall k : m \leq k \leq n$

$$(8.50) \quad 0 \leq \sum_{i=m}^k (-1)^{i-m} a_i \leq a_m$$

Case 3. Further restrict $n-m$ to odd integers. Since $\langle a_n \rangle$ is non-negative $a_{n+1} \geq 0$ along with the induction hypothesis

$$(8.51) \quad 0 \leq a_{n+1} + \sum_{i=m}^n (-1)^{i-m} a_i$$

$$(8.52) \quad = (-1)^{n+1-m} a_{n+1} + \sum_{i=m}^n (-1)^{i-m} a_i$$

$$(8.53) \quad = \sum_{i=m}^{n+1} (-1)^{i-m} a_i$$

Since $\langle a_n \rangle$ is decreasing

$$(8.54) \quad a_{n+1} \leq a_n \Rightarrow a_{n+1} - a_n \leq 0$$

Along with the induction hypothesis

$$(8.55) \quad a_m \geq a_{n+1} - a_n + \sum_{i=m}^{n-1} (-1)^{i-m} a_i$$

$$(8.56) \quad = (-1)^{n+1-m} a_{n+1} + (-1)^{n-m} a_n + \sum_{i=m}^{n-1} (-1)^{i-m} a_i$$

$$(8.57) \quad = \sum_{i=m}^{n+1} (-1)^{i-m} a_i$$

Case 4. Now restrict $n - m$ to only even integers. Since $\langle a_n \rangle$ is decreasing

$$(8.58) \quad a_n \geq a_{n+1} \Rightarrow a_n - a_{n+1} \geq 0$$

Applying the induction hypothesis

$$(8.59) \quad 0 \leq -a_{n+1} + a_n + \sum_{i=m}^{-1} (-1)^{i-m} a_i$$

$$(8.60) \quad = (-1)^{n+1-m} a_{n+1} + (-1)^{n-m} a_n + \sum_{i=m}^{n-1} (-1)^{i-m} a_i$$

$$(8.61) \quad = \sum_{i=m}^{n+1} (-1)^{i-m} a_i$$

Since $\langle a_n \rangle$ is non-negative $-a_{n+1} \leq 0$. Applying the hypothesis gives

$$(8.62) \quad a_m \geq -a_{n+1} + \sum_{i=m}^n (-1)^{i-m} a_i$$

$$(8.63) \quad = (-1)^{n+1-m} a_{n+1} + \sum_{i=m}^n (-1)^{i-m} a_i$$

$$(8.64) \quad = \sum_{i=m}^{n+1} (-1)^{i-m} a_i$$

Therefor for all n , $P(n) \Rightarrow P(n+1)$. By induction, for all n ,

$$(8.65) \quad 0 \leq \sum_{i=m}^n (-1)^{i-m} a_i \leq a_m$$

Since the sum is greater than zero

$$(8.66) \quad \sum_{i=m}^n (-1)^{i-m} a_i = \left| \sum_{i=m}^n (-1)^{i-m} a_i \right|$$

$$(8.67) \quad = \left| (-1)^m \sum_{i=m}^n (-1)^{i-m} a_i \right|$$

$$(8.68) \quad = \left| \sum_{i=m}^n (-1)^i a_i \right| \leq a_m$$

□

Theorem 8.15 (Alternating Series Test). *If a is decreasing, non-negative and converges to zero, then $\langle (-1)^n a_n \rangle$ is summable.*

Proof. Let $\langle s_n \rangle$ be a series of partial sums

$$(8.69) \quad s_n := \sum_{i=0}^n (-1)^i a_i$$

Fix $\varepsilon > 0$. Since $\lim_n a_n = 0$, choose some N such that if $n > N$

$$(8.70) \quad |a_n| < \varepsilon$$

Choose n and m such that $n > m > N$, then

$$(8.71) \quad \left| \sum_{i=m}^n (-1)^i a_i \right| \leq a_m \leq |a_m| < \varepsilon$$

By the fundamental criterion, $\langle a_n \rangle$ is summable. □

9. FUNCTIONAL LIMITS

Definition 9.1 (Limit). The *limit* L of a function $f : [a, b] \rightarrow \mathbb{R}$ at c is

$$(9.1) \quad \forall \varepsilon > 0 : \exists \delta > 0 : \forall x \in [A, B] : 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon$$

The limit at c of f is denoted by

$$(9.2) \quad \lim_{x \rightarrow c} f(x) = L$$

Theorem 9.2 (Sequential Limit Criterion).

$$(9.3) \quad \lim_{x \rightarrow c} f(x) = L$$

if and only if, for all sequences such that $\forall i : x_i \neq c$:

$$(9.4) \quad \lim_n x_n = c \Rightarrow \lim_n f(x_n) = L$$

Proof. Choose a sequence x such that $\forall x_n \neq c$ and

$$(9.5) \quad \lim_n x_n = c$$

Fix $\varepsilon > 0$. By assumption $\exists \delta > 0$ such that for every x

$$(9.6) \quad 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon$$

Since x_n converges to c , there is an N such that

$$(9.7) \quad n > N \Rightarrow |x_n - c| < \delta$$

Using the definition of the functional limit, this implies

$$(9.8) \quad |f(x_n) - L| < \varepsilon$$

Thus

$$(9.9) \quad \lim_n f(x_n) = L$$

For the converse, assume

$$(9.10) \quad \lim_{x \rightarrow c} f(x) \neq L$$

Then there is an $\varepsilon > 0$ such that for every $\delta > 0$ there is an x such that

$$(9.11) \quad 0 < |x - c| < \delta \wedge |f(x) - L| \geq \varepsilon$$

Choose such an ε . Generate a sequence such that

$$(9.12) \quad 0 < |x_n - c| < \frac{1}{n} \wedge |f(x_n) - L| \geq \varepsilon$$

By the squeeze theorem

$$(9.13) \quad \lim_n x_n = c$$

The other part of the generator

$$(9.14) \quad \forall n : |f(x_n) - L| \geq \varepsilon$$

implies

$$(9.15) \quad \lim_n f(x_n) \neq L$$

Thus

$$(9.16) \quad \lim_n x_n = c \Rightarrow \lim_n f(x_n) = L \Rightarrow \lim_{x \rightarrow c} f(x) = L$$

□

Theorem 9.3 (Squeeze Theorem). *If*

$$(9.17) \quad \lim_{x \rightarrow c} f(x) = L$$

$$(9.18) \quad \lim_{x \rightarrow c} g(x) = L$$

and

$$(9.19) \quad f(x) \leq h(x) \leq g(x)$$

Then

$$(9.20) \quad \lim_{x \rightarrow c} h(x) = L$$

Theorem 9.4 (Uniqueness). *If a function converges to a limit at some point, then that limit is unique.*

Theorem 9.5.

$$(9.21) \quad \lim_{x \rightarrow c} f(x) = F$$

$$(9.22) \quad \lim_{x \rightarrow c} g(x) = G$$

implies

$$(9.23) \quad \lim_{x \rightarrow c} [f(x) + g(x)] = F + G$$

Theorem 9.6.

$$(9.24) \quad \lim_{x \rightarrow c} f(x) = F$$

$$(9.25) \quad \lim_{x \rightarrow c} g(x) = G$$

implies

$$(9.26) \quad \lim_{x \rightarrow c} f(x)g(x) = FG$$

10. CONTINUOUS FUNCTIONS

Definition 10.1 (Continuous Function). A function f is continuous at a if

$$(10.1) \quad \lim_{x \rightarrow a} f(x) = f(a)$$

Theorem 10.2 (Sequential Limit Criterion). *For continuous functions f*

$$(10.2) \quad \lim_n f(x_n) = f\left(\lim_n x_n\right)$$

Theorem 10.3 (ε - δ Condition).

Theorem 10.4 (Closed Set Condition). *If $f : M \rightarrow N$ is continuous and M is closed, then the pre-image of each closed set in N is closed in M .*

Proof. Let I be a closed subset of N . Define a sequence $s : \mathbb{N} \rightarrow I \subset M$ such that

$$(10.3) \quad \lim_n s_n = L \in M$$

By sequential convergence condition

$$(10.4) \quad \lim_n f(s_n) = f(\lim_n s_n) = f(L)$$

Since the set I is closed

$$(10.5) \quad \lim_n f(s_n) \in I \Rightarrow f(L) \in I$$

So $L \in f^{-1}[I]$ which implies that $f^{-1}I$ is closed □

Theorem 10.5. *If $f : M \rightarrow N$ is continuous and I is compact in M , then the image of f restricted to I is compact*

Proof. Let $b : \mathbb{N} \rightarrow f[I]$ and define $a : \mathbb{N} \rightarrow I$ such that

$$(10.6) \quad f(a_n) = b_n$$

Since I is compact, a has a subsequential limit in I

$$(10.7) \quad L := \lim_i a_{n_i} \in I$$

Since f is continuous

$$(10.8) \quad \lim_i b_{n_i} = \lim_i f(a_{n_i}) = f(L) \in f[I]$$

There for the image of f is restricted to A is compact. □

Lemma 10.6.

$$(10.9) \quad a_{n+1} := \frac{a_n}{2} \Rightarrow a_n = \frac{1}{2^{n-1}} a_1$$

Proof. The induction hypothesis is that, $\forall n \in \mathbb{Z}^+$

$$(10.10) \quad a_n = \frac{1}{2^{n-1}} a_1$$

Case 1 (Base). The base case is trivially true

$$(10.11) \quad a_1 = \frac{1}{2^{1-1}} a_1$$

Case 2 (Induction). Assume without loss of generality

$$(10.12) \quad a_n = \frac{1}{2^{n-1}} a_1$$

From the opening assumption

$$(10.13) \quad a_{n+1} = \frac{a_n}{2}$$

$$(10.14) \quad = \frac{1}{2} \frac{1}{2^{n-1}} a_1$$

$$(10.15) \quad = \frac{1}{2^{n+1-1}} a_1$$

By induction, $\forall n \in \mathbb{Z}^+$

$$(10.16) \quad a_n = \frac{1}{2^{n-1}} a_1$$

□

Theorem 10.7 (Intermediate Value Theorem). *If $f : [A, B] \rightarrow \mathbb{R}$ is continuous and $f(A) \leq u \leq f(B)$ or $f(A) \geq u \geq f(B)$, then there exists $C \in [A, B] : f(C) = u$*

Proof. Define four sequences, a , b , c , and e with the following algorithm

$$(10.17) \quad a_1 := A$$

$$(10.18) \quad b_1 := B$$

$$(10.19) \quad c_n := \frac{1}{2}(a_n + b_n) e_n = b_n - a_n$$

If $f(c_n) < u$:

$$(10.20) \quad a_{n+1} = c_n$$

$$(10.21) \quad b_{n+1} = b_n$$

Else:

$$(10.22) \quad a_{n+1} = a_n$$

$$(10.23) \quad b_{n+1} = c_n$$

Using the algorithm, if $f(c_n) = u$, then there is a $C = c_n \in [A, B] : f(C) = u$, and we are done. For the rest of this proof, we will assume that $\forall n : f(c_n) \neq u$. Finishing off the algorithm, if $f(c_n) > u$

$$(10.24) \quad a_{n+1} = a_n$$

$$(10.25) \quad b_{n+1} = c_n = \frac{a_n + b_n}{2}$$

$$(10.26) \quad e_{n+1} = \frac{a_n + b_n}{2} - a_n = \frac{b_n - a_n}{2}$$

If $f(c_n) < u$

$$(10.27) \quad a_{n+1} = c_n = \frac{a_n + b_n}{2}$$

$$(10.28) \quad b_{n+1} = b_n$$

$$(10.29) \quad e_{n+1} = b_n - \frac{a_n + b_n}{2} = \frac{b_n - a_n}{2}$$

Since $\forall n : a_n < c_n < b_n$, a_n is increasing and b_n is decreasing. Since they are also bounded by (A, B) , by the monotone convergence theorem, they both converge. From the definition of e_n

$$(10.30) \quad e_n = b_n - a_n$$

So

$$(10.31) \quad e_{n+1} = \frac{e_n}{2}$$

By the lemma 12.6

$$(10.32) \quad e_n = \frac{1}{2^{n-1}} e_1 = \frac{1}{2^{n-1}} (B - A)$$

which converges to zero. Then $\lim_n b_n = \lim_n a_n$. Let $C := \lim_n a_n$, for continuous functions f

$$(10.33) \quad \lim_n f(a_n) = f(C) = \lim_n f(b_n)$$

By the squeeze theorem, since $\forall n : f(a_n) \leq u \leq f(b_n)$

$$(10.34) \quad f(C) = u$$

Thus it is proved that there exists such a C for $f(A) \leq u \leq f(B)$. If $f(A) \geq u \geq f(B)$, then

$$(10.35) \quad -f(A) \leq -u \leq -f(B)$$

Let $g(x) = -f(x)$ and $v = -u$, then

$$(10.36) \quad g(A) \leq v \leq g(B)$$

By the earlier result, $\exists C \in [A, B] : g(C) = v$, then $-f(C) = -u$ or $f(C) = u$. $\square$

Theorem 10.8. *If $f : M \rightarrow N$ is continuous and S is compact in M , then $f[S]$ is compact.*

Corollary 10.9 (Bounded Value Theorem). *If $f : M \rightarrow N$ is continuous and S is compact in M , then $f[S]$ is bounded.*

Theorem 10.10 (Bounded Value Theorem). *If $f : [a, b] \rightarrow \mathbb{R}$ is a continuous function, then f is bounded.*

$$(10.37) \quad \exists C : \forall x \in [a, b] : m \leq f(x) \leq M$$

Proof. Let $f : [A, B] \rightarrow \mathbb{R}$ be any continuous function. Assume without loss of generality

$$(10.38) \quad \forall C : \exists x \in [A, B] : f(x) > C$$

Since $\mathbb{Z}^+ \subset \mathbb{R}$

$$(10.39) \quad \forall n \in \mathbb{Z}^+, \exists x \in [A, B] : f(x) > n$$

Then there is a sequence x such that for all n

$$(10.40) \quad x_n \in [A, B] \wedge f(x_n) > n$$

By the point peak theorem, there is a monotonic subsequence $x \circ k$. Since x is bounded, the subsequence $x \circ k$ is bounded. Since $x \circ k$ is bounded and monotone

$x \circ k$ converges by the monotone convergence theorem. Since f is continuous, $f(x \circ k)$ also converges. Since $x \circ k$ is a subsequence

$$(10.41) \quad k_i \geq i$$

By trichotomy, for all i

$$(10.42) \quad f(x_{k_i}) > i$$

Thus $f(x \circ k)$ diverges, which is a contradiction. Therefore f is bounded above. $\square$

Theorem 10.11 (Extreme Value Theorem). *If $f : [A, B] \rightarrow \mathbb{R}$ is continuous*

$$(10.43) \quad \exists M \in [A, B], \forall x \in [A, B] : f(x) \leq f(M)$$

Proof. By the boundedness theorem, there is a bound, B to the continuous function. Define four sequences a, b, c, e such that

$$(10.44) \quad a_1 := f(A)$$

$$(10.45) \quad b_1 := B$$

For all n :

$$(10.46) \quad c_n := \frac{a_n + b_n}{2}$$

$$(10.47) \quad e_n := b_n - a_n$$

If c_n is a bound of the continuous function ($\forall x \in [A, B] : c_n \geq f(x)$):

$$(10.48) \quad a_{n+1} := a_n$$

$$(10.49) \quad b_{n+1} := c_n$$

Else

$$(10.50) \quad a_{n+1} := c_n$$

$$(10.51) \quad b_{n+1} := b_n$$

If c_n is a bound

$$(10.52) \quad a_{n+1} = a_n$$

$$(10.53) \quad b_{n+1} = c_n = \frac{a_n + b_n}{2}$$

$$(10.54) \quad e_{n+1} = \frac{a_n + b_n}{2} - a_n = \frac{b_n - a_n}{2}$$

If c_n is not a bound

$$(10.55) \quad a_{n+1} = c_n = \frac{a_n + b_n}{2}$$

$$(10.56) \quad b_{n+1} = b_n$$

$$(10.57) \quad e_{n+1} = b_n - \frac{a_n + b_n}{2} = \frac{b_n - a_n}{2}$$

Since $\forall n : a_n < c_n < b_n$, a_n is increasing and b_n is decreasing. Since they are also bounded by (A, B) , by the monotone convergence theorem, they both converge. From the definition of e_n

$$(10.58) \quad e_n = b_n - a_n$$

So

$$(10.59) \quad e_{n+1} = \frac{e_n}{2}$$

By the lemma 12.6

$$(10.60) \quad e_n = \frac{1}{2^{n-1}} e_1 = \frac{1}{2^{n-1}} (B - A)$$

which converges to zero. Therefor

$$(10.61) \quad \lim_n (b_n - a_n) = 0$$

$$(10.62) \quad L := \lim_n b_n = \lim_n a_n$$

□

11. CONNECTED SPACES

Definition 11.1 (Connected Space). A set is *connected*, if it has no non-empty, proper, clopen subset. A set that is not connected is *disconnected*.

Theorem 11.2. *If $f : M \rightarrow N$ is continuous, and M is connected, then $f[M]$ is connected.*

Proof. Suppose $f[M]$ is disconnected. Then there is some non-empty, proper, clopen subset A .

$$(11.1) \quad A \subsetneq f[M]$$

The pre-image of A , then is a non-empty, proper subset of M

$$(11.2) \quad f^{-1}[A] \subsetneq M$$

By the open and closed set conditions, the preimage $f^{-1}[A]$ is clopen. But by assumption, there is no non-empty, proper, clopen subset. So $f[M]$ must be connected. □

Lemma 11.3. *If I is a connected interval on $\mathbb{R}$,*

$$(11.3) \quad \forall a, b \in I : a < b \Rightarrow \exists c \in I : a < c < b$$

Proof. Suppose its not,

$$(11.4) \quad \exists a, b \in I : a < b \wedge c \in (-\infty, a] \cup [b, \infty)$$

Let $J := (-\infty, a] \cup [b, \infty)$. The subsets of J , $(-\infty, a]$ and its conjugate $[b, \infty)$ are both closed, therefor both are clopen, and I is disconnected. □

Theorem 11.4 (Generalized Intermediate Value Theorem). *If $f : M \rightarrow \mathbb{R}$ is continuous and $I \subset M$ is connected, then*

$$(11.5) \quad \begin{aligned} & \forall a, b \in I : f(a) < f(b) \\ & \Rightarrow \exists c \in I : f(a) < f(c) < f(b) \end{aligned}$$

Proof. Since I is connected, $f[I]$ is connected. Hence there is a $\gamma \in f[I]$

$$(11.6) \quad f(a) < \gamma < f(b)$$

Therefor there is a $c \in I$ such that

$$(11.7) \quad \exists c \in I : f(a) < f(c) < f(b)$$

□

12. DIFFERENTIABLE FUNCTIONS

Definition 12.1 (Differentiable Function). A function f is *differentiable* at a if the following exists.

$$(12.1) \quad f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

This limit $f'(a)$ is called the *derivative* of f at a

Theorem 12.2. *Every differentiable function is continuous.*

Proof.

$$(12.2) \quad f(x) - f(a) = \frac{f(x) - f(a)}{x - a}(x - a)$$

$$(12.3) \quad \lim_{x \rightarrow a} [f(x) - f(a)] = \lim_{x \rightarrow a} \left[\frac{f(x) - f(a)}{x - a}(x - a) \right]$$

If $f : [A, B] \rightarrow \mathbb{R}$ is differentiable at $a \in [A, B]$, then $f'(a)$ exists.

$$(12.4) \quad f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$(12.5) \quad \begin{aligned} \lim_{x \rightarrow a} [f(x) - f(a)] &= f'(a) \cdot 0 \\ &= 0 \end{aligned}$$

$$(12.6) \quad \begin{aligned} \lim_{x \rightarrow a} f(x) &= \lim_{x \rightarrow a} f(a) \\ \lim_{x \rightarrow a} f(x) &= f(a) \end{aligned}$$

Thus $f(x)$ is continuous at a □

Theorem 12.3 (Chain Rule). *If $f : [a, b] \rightarrow I$ and $g : I \rightarrow \mathbb{R}$ are differentiable, then*

$$(12.7) \quad (f \circ g)'(x) = (f' \circ g)(x)g'(x)$$

Proof. Define u, v such that

$$(12.8) \quad u(t) := f' \circ g(x) - \frac{(f \circ g)(x) - (f \circ g)(t)}{g(x) - g(t)}$$

$$(12.9) \quad v(t) := g'(x) - \frac{g(x) - g(t)}{x - t}$$

The limit of u at x is then

$$(12.10) \quad \lim_{t \rightarrow x} u(t) = (f' \circ g)(x) - \lim_{t \rightarrow x} \frac{(f \circ g)(x) - (f \circ g)(t)}{g(x) - g(t)}$$

$$(12.11) \quad = (f' \circ g)(x) - \lim_{g(t) \rightarrow g(x)} \frac{(f \circ g)(x) - (f \circ g)(t)}{g(x) - g(t)}$$

$$(12.12) \quad = (f' \circ g)(x) - (f' \circ g)(x) = 0$$

And the limit of v at x is also

$$(12.13) \quad \lim_{t \rightarrow x} v(t) = g'(x) - \lim_{t \rightarrow x} \frac{g(x) - g(t)}{x - t}$$

$$(12.14) \quad = g'(x) - g'(x) = 0$$

Rearranging the definition of v gives

$$(12.15) \quad g(x) - g(t) = (x - t) [g'(x) - v(t)]$$

and the definition of u

$$(12.16) \quad (f \circ g)(x) - (f \circ g)(t) = [g(x) - g(t)] [f' \circ g(x) - u(t)]$$

Substituting gives

$$(12.17) \quad (f \circ g)(x) - (f \circ g)(t) = (x - t) [g'(x) - v(t)] [f' \circ g(x) - u(t)]$$

The definition of the limit of the composite function gives

$$(12.18) \quad (f \circ g)'(x) = \lim_{t \rightarrow x} \frac{(f \circ g)(x) - (f \circ g)(t)}{x - t}$$

$$(12.19) \quad = \lim_{t \rightarrow x} \frac{x - t}{x - t} [g'(x) - v(t)] [f' \circ g(x) - u(t)]$$

$$(12.20) \quad = (f' \circ g)(x) \cdot g'(x)$$

□

Lemma 12.4. *If f is continuous on $[a, b]$ and differentiable on (a, b) , then*

$$(12.21) \quad f(a) = f(b) \Rightarrow \exists c \in (a, b) : f'(c) = 0$$

Proof. Since f is continuous on $[a, b]$, f achieves a maximum $M \in [a, b]$ and a minimum $m \in [a, b]$.

Case 1. Suppose m and M are both end points of $[a, b]$. Then

$$(12.22) \quad f(a) = f(b) \Rightarrow f(m) = f(M)$$

Since $f(m)$ and $f(M)$ are minimum and maximums of f respectively

$$(12.23) \quad \forall x \in [a, b] : f(m) \leq f(x) \leq f(M)$$

along with $f(m) = f(M)$, implies

$$(12.24) \quad \forall x \in [a, b] : f(m) = f(x) = f(M)$$

So f is a constant function on $[a, b]$, and the derivative of the constant function is

$$(12.25) \quad \forall x \in (a, b) : f'(x) = 0$$

Case 2. Now suppose that m or M is not an end point of $[a, b]$. Then there is a $c \in (a, b)$ such that f has a local extrema at c . Since the derivative at a local extrema is zero.

$$(12.26) \quad f'(c) = 0$$

So for all cases,

$$(12.27) \quad \exists c \in (a, b) : f'(c) = 0$$

□

Theorem 12.5 (Generalized Mean Value Theorem). *If f and g are continuous on $[a, b]$ and differentiable on (a, b) , then*

$$(12.28) \quad \exists c \in (a, b) : [f(b) - f(a)] g'(c) = [g(b) - g(a)] f'(c)$$

Proof. Define h such that

$$(12.29) \quad h(x) := [f(b) - f(a)]g(x) - [g(b) - g(a)]f(x)$$

Since f and g both are continuous on $[a, b]$ and differentiable on (a, b) , h is continuous on $[a, b]$ and differentiable on (a, b) . So the derivative of h on (a, b) is

$$(12.30) \quad h'(x) = [f(b) - f(a)]g'(x) - [g(b) - g(a)]f'(x)$$

At the end points

$$\begin{aligned} h(a) &= f(b)g(a) - f(a)g(a) - g(b)f(a) + g(a)f(a) \\ &= f(b)g(a) - f(a)g(b) \end{aligned}$$

And

$$\begin{aligned} h(b) &= f(b)g(b) - f(a)g(b) - g(b)f(b) + g(a)f(b) \\ (12.31) \quad &= f(b)g(a) - f(a)g(b) \end{aligned}$$

So

$$(12.32) \quad h(a) = h(b)$$

By the above lemma

$$(12.33) \quad \exists c \in (a, b) : h'(c) = 0$$

So there is a $c \in (a, b)$ such that

$$(12.34) \quad h'(c) = 0 = [f(b) - f(a)]g'(c) - [g(b) - g(a)]f'(c)$$

or

$$(12.35) \quad [f(b) - f(a)]g'(c) = [g(b) - g(a)]f'(c)$$

□

Theorem 12.6 (L'hôpital's Rule). *If f and g are differentiable functions and*

$$(12.36) \quad \begin{aligned} \lim_{x \rightarrow a} f(x) &= 0 \\ \lim_{x \rightarrow a} g(x) &= 0 \end{aligned}$$

Then

$$(12.37) \quad \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

Proof. Choose some ε and ε' such that $\varepsilon > \varepsilon' > 0$. By 14.37, there is some $\delta > 0$ such that $\forall x \in [a, b]$

$$(12.38) \quad 0 < |x - c| < \delta \Rightarrow \left| \frac{f'(x)}{g'(x)} - L \right| < \varepsilon'$$

For all $(x, y) \subset (c, c + \delta)$ or $(x, y) \subset (c - \delta, c)$, by generalized mean value theorem, there is a $t \in (x, y)$ such that

$$(12.39) \quad \frac{f(x) - f(y)}{g(x) - g(y)} = \frac{f'(t)}{g'(t)}$$

Hence

$$\begin{aligned}
 & \left| \frac{f(x) - f(y)}{g(x) - g(y)} - L \right| < \varepsilon' \\
 & L - \varepsilon' < \frac{f(x) - f(y)}{g(x) - g(y)} < L + \varepsilon' \\
 (12.40) \quad & L - \varepsilon' \leq \lim_{x \rightarrow c} \frac{f(x) - f(y)}{g(x) - g(y)} \leq L + \varepsilon'
 \end{aligned}$$

Assume that $\lim_{x \rightarrow c} f(x) = 0$ and $\lim_{x \rightarrow c} g(x) = 0$, then

$$(12.41) \quad L - \varepsilon' \leq \frac{f(y)}{g(y)} \leq L + \varepsilon'$$

$$(12.42) \quad \left| \frac{f(y)}{g(y)} - L \right| \leq \varepsilon' < \varepsilon$$

So for every $\varepsilon > 0$, there is a $\delta > 0$ such that for all $y \in [a, b]$

$$(12.43) \quad |y - c| < \delta \Rightarrow \left| \frac{f(y)}{g(y)} - L \right| < \varepsilon$$

Or

$$(12.44) \quad \lim_{y \rightarrow c} \frac{f(y)}{g(y)} = L$$

□

13. INTEGRABLE FUNCTIONS

Definition 13.1 (Partition). A *partition* on $[a, b]$ is a finite, increasing sequence $x : [0, n] \rightarrow [a, b]$, such that

$$\begin{aligned}
 & x_0 = a \\
 (13.1) \quad & x_n = b
 \end{aligned}$$

The sequence of lengths of partitions Δx_i is defined simply as

$$(13.2) \quad \Delta x_i := x_{i-1} - x_i$$

The *mesh* of a partition is given by

$$(13.3) \quad \text{mesh } x := \max_i \Delta x_i$$

A sequence of sample points $t : [0, n] \rightarrow [a, b]$ for a partition

$$(13.4) \quad \forall i : x_{i-1} \leq t_i \leq x_i$$

A *partition pair* is a tuple of sequences, the partition x and one of its sequence of sample points t .

Definition 13.2 (Partition Sum). The *partition sum* of a function on f with partition pair x, t is given by

$$(13.5) \quad R(f, x, t) := \sum_{i=1}^n f(t_i) \Delta x_i$$

Definition 13.3 (Integral). A function is *integrable* on $[a, b]$ if there is an $I \in \mathbb{R}$ such that, for all $\varepsilon > 0$ there is a $\delta > 0$ such that for all partitions x

$$(13.6) \quad \text{mesh } x < \delta \Rightarrow |R(f, x, t) - I| < \varepsilon$$

If f is integrable on $[a, b]$, I is the integral of f over $[a, b]$ and it is denoted by

$$(13.7) \quad \int_a^b f(x) \, dx = I = \lim_{\text{mesh } x \rightarrow 0} \sum_{i=1}^n f(t_i) \Delta x_i$$

Definition 13.4 (Upper Sum).

$$(13.8) \quad U(f, x) := \sum_{i=1}^n M_i \Delta x_i$$

where

$$(13.9) \quad M_i := \sup\{f(x) : t_{i-1} \leq x_i \leq t_i\}$$

Definition 13.5 (Lower Sum).

$$(13.10) \quad L(f, x) := \sum_{i=1}^n m_i \Delta x_i$$

where

$$(13.11) \quad m_i := \inf\{f(x) : t_{i-1} \leq x_i \leq t_i\}$$

Lemma 13.6. If t^* is a refinement of t , such that $t = t^* \circ p$

$$(13.12) \quad \sum_{j=p_{i-1}+1}^{p_i} a(t_j^* - t_{j-1}^*) = a(t_i - t_{i-1})$$

Proof.

$$(13.13) \quad \sum_{j=p_{i-1}+1}^{p_i} a(t_j^* - t_{j-1}^*) = a \sum_{j=p_{i-1}+1}^{p_i} (t_j^* - t_{j-1}^*)$$

$$(13.14) \quad = a \sum_{j=i-1+1}^i (t_{p_j}^* - t_{p_{j-1}}^*)$$

$$(13.15) \quad = a \sum_{j=i}^i (t_j - t_{j-1})$$

$$(13.16) \quad = a(t_i - t_{i-1})$$

□

Theorem 13.7. If x^* is a refinement of x , then

$$(13.17) \quad \begin{aligned} L(f, x) &\leq L(f, x^*) \\ U(f, x) &\geq U(f, x^*) \end{aligned}$$

Proof. Define M , M^* , m and m^* such that

$$\begin{aligned} M_i &= \sup\{f(x) : t_{i-1} \leq x \leq t_i\} \\ M_i^* &= \sup\{f(x) : t_{i-1}^* \leq x \leq t_i^*\} \\ m_i &= \inf\{f(x) : t_{i-1} \leq x \leq t_i\} \\ m_i^* &= \inf\{f(x) : t_{i-1}^* \leq x \leq t_i^*\} \end{aligned}$$

Let $p : [0, n] \rightarrow [0, n^*]$ be a finite sequence such that

$$(13.18) \quad t = t^* \circ p$$

Then M and m can be written as

$$\begin{aligned} M_i &= \sup\{f(x) : t_{p_{i-1}}^* \leq x \leq t_{p_i}^*\} \\ m_i &= \inf\{f(x) : t_{p_{i-1}}^* \leq x \leq t_{p_i}^*\} \end{aligned}$$

Thus $\forall i \in [0, n] : \forall j \in [p_{i-1}, p_i] : m_j^* \geq m_i$ and $M_j^* \leq M_i$, thus

$$(13.19) \quad \sum_{j=p_{i-1}+1}^{p_i} m_j^*(t_j^* - t_{j-1}^*) \geq \sum_{j=p_{i-1}+1}^{p_i} m_i(t_j^* - t_{j-1}^*)$$

$$(13.20) \quad \sum_{j=p_{i-1}+1}^{p_i} M_j^*(t_j^* - t_{j-1}^*) \leq \sum_{j=p_{i-1}+1}^{p_i} M_i(t_j^* - t_{j-1}^*)$$

By the above lemma

$$(13.21) \quad \sum_{j=p_{i-1}+1}^{p_i} m_j^*(t_j^* - t_{j-1}^*) \geq m_i(t_i - t_{i-1})$$

$$(13.22) \quad \sum_{j=p_{i-1}+1}^{p_i} M_j^*(t_j^* - t_{j-1}^*) \leq M_i(t_i - t_{i-1})$$

And

$$(13.23) \quad \sum_{i=1}^n \sum_{j=p_{i-1}+1}^{p_i} m_j^*(t_j^* - t_{j-1}^*) \geq \sum_{i=1}^n m_i(t_i - t_{i-1})$$

$$(13.24) \quad \sum_{i=1}^n \sum_{j=p_{i-1}+1}^{p_i} M_j^*(t_j^* - t_{j-1}^*) \leq \sum_{i=1}^n M_i(t_i - t_{i-1})$$

And finally

$$(13.25) \quad \sum_{i=1}^{n^*=p_n} m_i^*(t_i^* - t_{i-1}^*) \geq \sum_{i=1}^n m_i(t_i - t_{i-1})$$

$$(13.26) \quad \sum_{i=1}^{n^*=p_n} M_i^*(t_i^* - t_{i-1}^*) \leq \sum_{i=1}^n M_i(t_i - t_{i-1})$$

Giving

$$(13.27) \quad L(f, t^*) \geq L(f, t)$$

$$(13.28) \quad U(f, t^*) \leq U(f, t)$$

□

Lemma 13.8. *For a partition P*

$$(13.29) \quad L(f, P) \leq U(f, P)$$

Proof. Define M and m such that

$$(13.30) \quad M_i = \sup\{f(x) : t_{i-1} \leq x \leq t_i\}$$

$$(13.31) \quad m_i = \inf\{f(x) : t_{i-1} \leq x \leq t_i\}$$

then for every $i : m_i \leq M_i$, thus

$$(13.32) \quad \sum_{i=1}^n m_i(t_i - t_{i-1}) \leq \sum_{i=1}^n M_i(t_i - t_{i-1})$$

Hence $L(f, t) \leq U(f, t)$ □

Theorem 13.9. *For all partitions, P_1, P_2 ,*

$$(13.33) \quad L(f, P_1) \leq U(f, P_2)$$

Proof. Let t^* be the common refinement of t_1 and t_2 , then

$$(13.34) \quad L(f, t_1) \leq L(f, t^*)$$

$$(13.35) \quad U(f, t^*) \leq U(f, t_2)$$

By the above lemma

$$(13.36) \quad L(f, t^*) \leq U(f, t^*)$$

Hence

$$(13.37) \quad L(f, t_1) \leq U(f, t_2)$$
□

Theorem 13.10. *A bounded function f is integrable on $[a, b]$, if for every $\varepsilon > 0$, there is a partition x such that*

$$(13.38) \quad U(f, x) - L(f, x) < \varepsilon$$

Proof. Let $\varepsilon > 0$ and suppose such a partition x exists such that

$$(13.39) \quad U(f, x) - L(f, x) < \varepsilon$$
□

14. PRIMITIVES

Definition 14.1 (Primitive). Let f be continuous on (a, b) . The function F is a primitive of f , if F is differentiable on (a, b) and

$$(14.1) \quad \forall x \in (a, b) : F'(x) = f(x)$$

Lemma 14.2. *If f is continuous on $[a, b]$, then the function $F(x) = \int_a^x f(t) dt$ is uniformly continuous on $[a, b]$.*

Lemma 14.3. *If f is continuous on $[a, b]$, then the function $F(x) = \int_a^x f(t) dt$ is differentiable on (a, b) .*

Theorem 14.4. *If f is continuous on $[a, b]$, then the function $F(x) = \int_a^x f(t) dt$ is a primitive of f .*

Theorem 14.5. *If f is integrable on $[a, b]$, F is differentiable on $[a, b]$, and F is a primitive of f , then*

$$(14.2) \quad \int_a^b f(x) \, dx = F(b) - F(a)$$